👍Confidently Incorrect👎: Do RMs Reward Making Stuff Up?

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Our Website

Introduction

RLHF aligns an LLM using an RM trained on human feedback data.

- Human annotators naturally care about factuality
- So *why do LLMs still hallucinate?*

RLFR is a recent RL pipeline using factuality probes as reward scores instead.

- Why are these probes more effective as rewards for reducing hallucinations?

Methods

We analyze **ArmoRM,** a MoE RM with 19 interpretable scores, and hallucination probes used in **RLFR**.

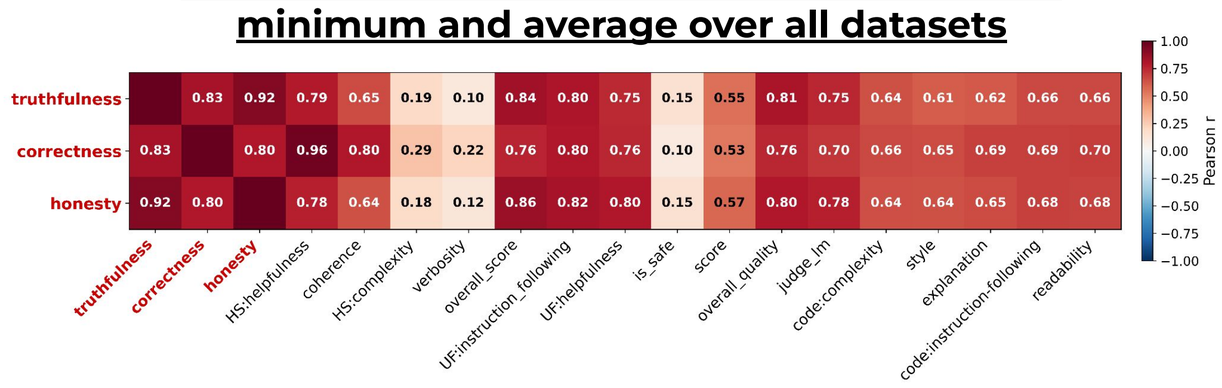
- We compare AUROC on 3 QA datasets and 2 human feedback datasets. *(examples on website)*
  - RQ1: Are factual features correlated with other features like complexity?
- We train an SAE on ArmoRM and compare its features with ArmoRM's objectives
  - RQ2: How do SAE features relate to ArmoRM's objectives and human annotations?

Do RMs conflate factuality with other features?

- Measure correlation of rewards on responses from off-policy models.
- Calculate the AUROC of ArmoRM objectives

(total: 29,624 evaluation data points)

Unrelated reward scores show correlation for minimum and average over all datasets

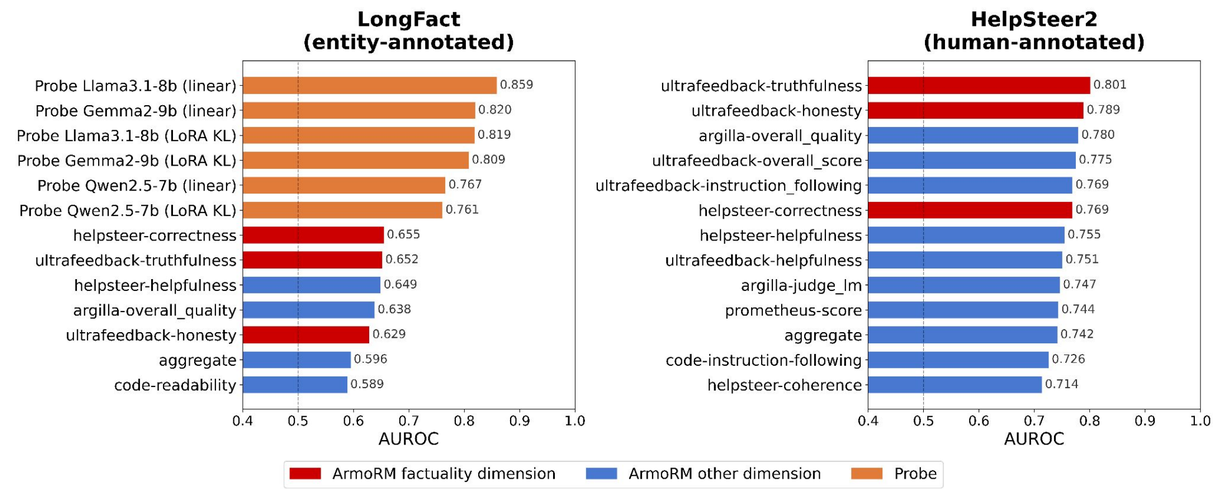


A response deemed more helpful/detailed may be erroneously interpreted as more factual

RQ1 Conclusion:

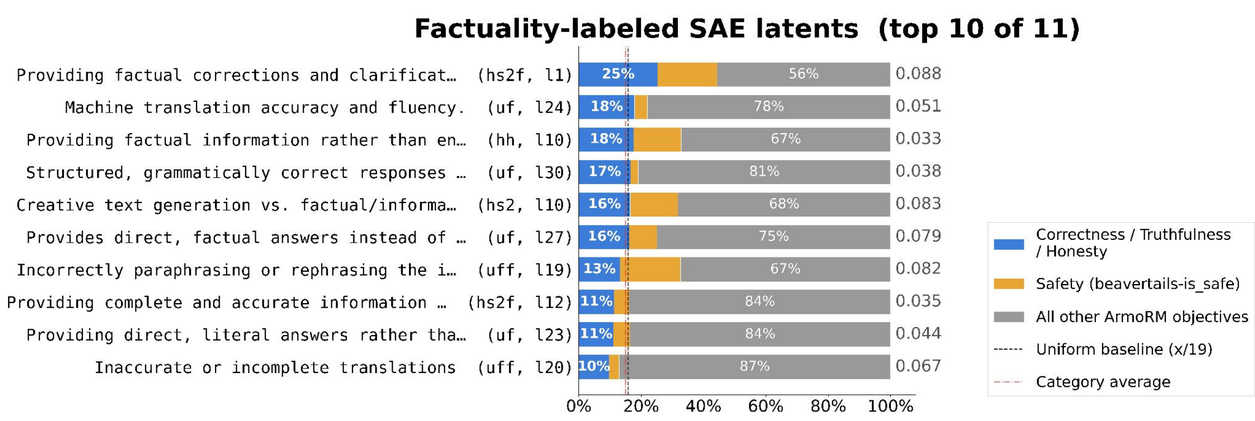
Unrelated features may cause high factuality scores. ArmoRM's helpfulness score is a semi-reliable classifier of factuality.
On entity-level datasets, hallucination probes > ArmoRM. This may be why RLFR is more effective.

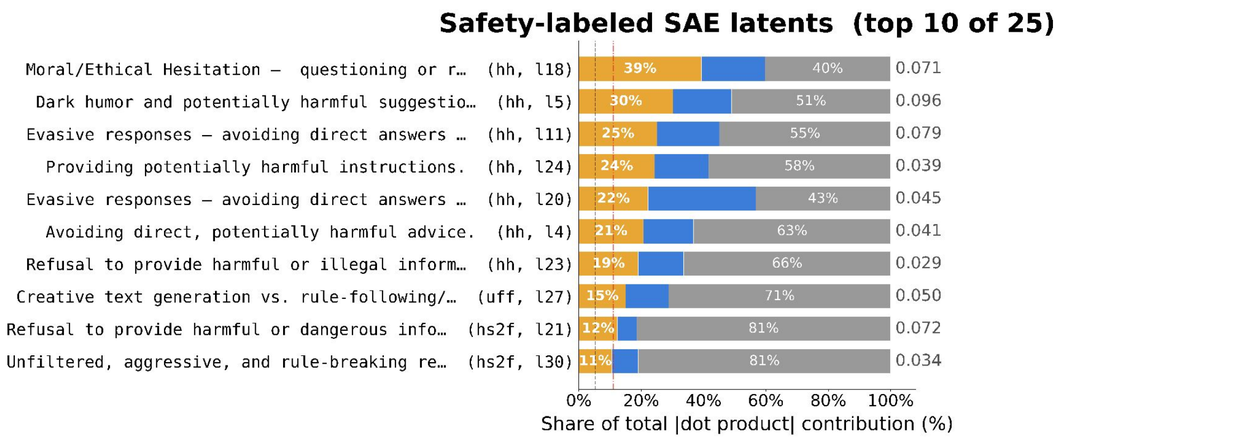
AUROC of non-factuality dimensions are on-par with factuality dimensions on LongFact



⭐Hallucination probes > ArmoRM on entity-annotated QA.
⭐ArmoRM > Probes on human factuality annotations.
⭐ Helpfulness should be independent from factuality but can be a reliable classifier

Factuality does not correlate with its ArmoRM dimension, while other features do!





- Safety features contribute 2 x baseline to a single is_safe reward.
- Factuality features don't, scoring slightly below random baseline.
- Note: each set of features score random on the other reward set.

How do SAE features compare to ArmoRM's reward heads?

- Train a Matryoshka SAE (k=8, d=32) to reconstruct the representation diff on the final layer of ArmoRM
- Use gemma-2-9b-it on contrastive examples to label and categorize features
- Use dot product of SAE decoder and ArmoRM reward head to attribute contribution to ArmoRM directions

RQ2 Conclusion:

Safety-labeled SAE features strongly align with their corresponding ArmoRM direction, implying autointerpretability can succeed here.

Factuality-labeled features do not, suggesting ArmoRM may not effectively utilize all factuality-related features.

Overall Conclusion:

RMs trained on human feedback data may exhibit biases towards coherent or helpful-sounding completions.

RLFR uses a less-biased signal of factuality to better reward models.

Factuality latents are not effectively utilized by ArmoRM reward

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